

Lab 1 – VibeQueue

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1. **Introduction**

The hospitality industry is a broad business sector focused on customer satisfaction, food and beverage service, travel, and more. One of the first things customers notice when entering a facility is the atmosphere. An establishment's atmosphere directly affects how long customers stay, customer satisfaction, and per-visit spending. (Milliman, 1982; North et al., 1999) Music genre and tempo are among the top controllable environmental factors in food service settings.

About 1 million independent food and beverage locations operate in the U.S. (National Restaurant Association [NRA], 2023). As a result, most small venues (<50 seats) cannot afford proprietary music management hardware (\$500–\$1,500+ upfront cost). This creates multiple issues, including staff interruptions and baristas fielding 10–15 verbal song requests during the morning rush, which reduces service efficiency. Owners also lose control of the intended atmosphere when staff manually honor off-brand requests. This is common in bars with jukeboxes, where people pay to have a song played and may complain or try to override the music selection. An estimated ~600,000 independent U.S. cafes and restaurants in the under-50-seat segment represent an underserved addressable market (NRA, 2023) that needs a different music option.

Companies already provide customer-controlled music services, with benefits and drawbacks. TouchTunes (TouchTunes, n.d.) is a digital jukebox network found in many social venues across America today. Their proprietary jukebox hardware requires a \$500–\$1,500+ upfront cost and is designed for large bars, restaurants, and establishments. While TouchTunes is one of the most recognized jukebox options, it still comes with its problems. The cost and design are inaccessible for small businesses. Also, it has a limited customer-facing digital interface and

is staff-mediated only. Rockbot (Rockbot, n.d.) is also another digital jukebox company that targets large to medium businesses. While their hardware and software may not be as widespread as TouchTunes, they charge \$25 a month, along with either a \$149 or \$380 one-time fee (Rockbot, 2026). However, while their prices seem low, the add-ons they include for businesses can add up over time if companies want to give their customers the best experience. The last option would be manual staff requests or generic personal streaming, yet even those have their own discrepancies. With customers verbally requesting songs, it could interrupt the workflow or lead to inconsistent outcomes, and off-brand requests could compromise the venue's atmosphere. On the other hand, generic personal requests are illegal for commercial use without a commercial license (Spotify, 2023), and the company can lose out on customer interaction and the control of the atmosphere.

Overall, no real-time, affordable, staff-free, legally compliant music management tool exists for small venues. Customers have no digital channel to influence venue atmosphere, and owners lack a secondary revenue stream from music engagement. The ~600,000 under-50-seat U.S. venues represent this gap, with no viable current solution (NRA, 2023). What's needed is a low-cost solution viable on a small-business budget (target: <\$50/month subscription), with no proprietary hardware requirement and customer-facing interaction without interrupting staff. Legal compliance with commercial music licensing (ASCAP, BMI, SESAC) is also needed to prevent licensing disputes. That's why there is VibeQueue, "Your Vibes, Managed", a web-based, hardware-agnostic, subscription jukebox platform for local spaces. It provides a cheap & easy option to better manage your environment's atmosphere with owner/staff final song-selection features, while giving guests the freedom to listen to what they want.

2. **Product Description**

VibeQueue is a software-only solution that replaces manual staff music-request processes with an integrated digital music ecosystem for businesses. The system lets customers influence the music played at a venue through their personal smartphones while giving owners and staff full control over the available song pool and music queue. VibeQueue also offers an optional pay-to-request feature that lets businesses generate a secondary revenue stream through customer song requests. To support legal commercial music playback, VibeQueue integrates with SoundMachine, which provides commercially licensed music playback rather than relying on customers' personal music-streaming accounts.

VibeQueue differs from traditional music-request systems because it is entirely software-based and can run on existing venue equipment. The system uses a subscription-based model with a low monthly cost instead of obliging businesses to make a large upfront investment in specialized hardware. Customers interact directly with the system through a QR-code interface instead of interrupting employees with verbal requests. VibeQueue also addresses legally compliant commercial music playback by using SoundMachine and its applicable ASCAP, BMI, and SESAC licensing, rather than relying on unlicensed personal streaming services.

2.1. Key Product Features and Capabilities

The Vibe Engine is one of *the main components* that makes VibeQueue unique. It is a proprietary validation system that evaluates requested songs based on venue-defined characteristics such as genre, tempo, mood, and explicit content. The Vibe Engine uses information obtained from Soundcharts and compares it against the rules stored in the Venue Database before approving or rejecting a request. This lets businesses maintain their desired atmosphere automatically, rather than requiring employees to review every song request. The Business Control Center gives owners and staff an integrated dashboard to manage the music queue, configure venue constraints, monitor requests, and review analytics.

VibeQueue also includes Dynamic Vibe Moderation, which automatically enforces the rules the business sets throughout the music-request process. For example, a café could allow acoustic, indie, and lo-fi music while blocking explicit content and songs above a specific tempo. The system would automatically prevent requests that violate those requirements. An optional Pay-to-Request feature lets customers boost a song in the queue for about \$0.99–\$1.99, with VibeQueue receiving a 5% platform revenue share. Finally, the Licensed Streaming feature uses SoundMachine to provide commercially licensed audio playback.

2.2. Major Components (Hardware/Software)

VibeQueue is a software-only solution and does not require proprietary or dedicated hardware. The system is designed to run on equipment a business already owns, such as a tablet, smartphone, laptop, and existing speaker system. This hardware-agnostic design reduces

deployment cost and complexity because businesses do not need to purchase specialized equipment to use the system. The Soundcharts API provides the music metadata the Vibe Engine needs to evaluate requests. This information can include characteristics such as genre, tempo, mood, popularity, and explicit-content indicators. Once a request is approved, SoundMachine provides licensed commercial music playback. The Local Server supports communication between the Vibe Engine and SoundMachine and handles audio routing and playback status. The Venue Database stores information such as the venue's vibe settings, music queue state, and transaction and revenue data.

The Business Control Center Dashboard provides owners and staff with a real-time management interface for the system. Through the dashboard, authorized users can manage the music queue, configure venue restrictions, monitor customer requests, and review available analytics. Together, these components create a system in which customers can submit requests through their smartphones, the Vibe Engine can automatically validate those requests, approved songs can be added to the queue, and SoundMachine can provide the final licensed audio playback through the venue's existing speaker system.

VibeQueue will use the LAMP technology stack, consisting of Linux, Apache, MySQL, and PHP, as the primary development environment for the application. The system will also integrate with the Soundcharts API to obtain music analytics and metadata required by the Vibe Engine. Soundcharts information such as genre, tempo, mood, popularity, and explicit-content status will allow the system to determine whether a customer's requested song satisfies the venue's requirements. The VibeQueue system will integrate with the SoundMachine API to provide licensed commercial audio streaming. A QR code generation library will create venue-

specific QR codes that direct customers to the appropriate VibeQueue portal. The front end will use a mobile-first responsive web design so that customers can access the system from smartphones and other internet-connected devices without installing a dedicated application.

3. Identification of Case Study

VibeQueue is being developed for small, independent businesses that play music for customers and want to provide greater customer interaction without requiring employees to manage song requests manually. The primary case study is Brewpoint Café, a single-location independent coffee shop with approximately 30 seats. Brewpoint Café was selected because its staff frequently experience interruptions from customer music requests, while the owner must also maintain control over the café's intended atmosphere. The case study will allow the development team to evaluate the VibeQueue prototype with a small group of users and collect feedback regarding the system's usability, functionality, and effectiveness. Brewpoint Café currently experiences frequent interruptions when customers verbally request songs from baristas. During the morning rush, baristas may be interrupted about 10–15 times as customers request specific songs. In addition, the current music-request process does not provide the business with an opportunity to generate revenue from customer participation. VibeQueue solves these problems by letting customers submit requests digitally, automatically enforcing the owner's music restrictions, and offering an optional paid song-boosting feature. Although Brewpoint Café is the initial case study, VibeQueue could eventually be expanded to additional

businesses and markets. The system could support small café chains with approximately 5–10 locations *by providing a centralized* Business Control Center and individual vibe profiles for each location. The product could also be adapted for college dining halls, hotel lobbies, fitness studios, barbershops, and other businesses where background music contributes to the customer experience. In addition, VibeQueue could be offered as a B2B SaaS product for venue-management companies that oversee multiple businesses.

5. Glossary

API (Application Programming Interface): set of protocols allowing software applications to communicate with each other

ASCAP (American Society of Composers, Authors and Publishers): U.S. performing rights organization licensing music for commercial use

BMI (Broadcast Music Inc.) :U.S. performing rights organization managing commercial music licensing on behalf of songwriters and publishers

Business Control Center :VibeQueue’s owner-facing web dashboard for configuring vibe constraints, managing the song queue, and viewing analytics

Hardware-Agnostic: software designed to run on any existing device without requiring specific or proprietary hardware

LAMP:open-source web development stack: Linux (OS), Apache (web server), MySQL (database), PHP (scripting language)

MFCD (Major Functional Component Diagram):visual representation of a system’s major components and data flows between them (CS 410 definition)

QR Code (Quick Response Code): two-dimensional barcode scanned by smartphone camera to open a URL or application

SESAC (Society of European Stage Authors and Composers): U.S. performing rights organization providing commercial music licensing

SoundMachine: third-party commercial music streaming service holding ASCAP/BMI/SESAC licenses; used by VibeQueue for legal audio playback in venues

Soundcharts API: third-party music analytics API providing commercially safe song metadata: genre, tempo, mood, popularity, and explicit content flags

Vibe Engine: VibeQueue's proprietary algorithm evaluating customer song requests against venue-configured constraints before queuing them

6.

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